

Stochastic Methods in Finance

Assignment

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1 Goals and conditions

The goal of this assignment is to price a European floating-strike Asian call option by building and implementing a multi-period binomial tree model. Additionally, an approximation using the normal distribution will be developed to validate and speed up calculations. A real-world stock will be used to calibrate the model parameters.

The following conditions must be satisfied:

- (i) The assignment may be conducted in groups of up to four students. The group members must be announced to Ms. Barbara Langenegger no later than **Friday, 1 May 2026**. No further changes to the group composition will be accepted, unless a member decides to leave her or his group. In that case, the change must be announced by the leaving member via email to Ms. Barbara Langenegger, with all other group members in CC. The list of groups will be published on Canvas on 4 May 2026. The members of each group must be clearly indicated in the final solution document.
- (ii) The **deadline for the assignment is Monday, 18 May 2026 at 4:00 pm**. The solutions (as a single PDF file, or a ZIP file if multiple files) must be sent via email to Ms. Barbara Langenegger. She will confirm receipt via email. **Solutions submitted after the deadline will not be graded**. Please use the subject line "6,252,1 Stochastic Methods in Finance: Assignment" in your email.
- (iii) Present your analysis in an organized manner. Results must be reported (tables and figures) and **commented on** in the main document. I will not run any code on my computer to obtain the results of your analysis! All formulas used to estimate model parameters or risk measures must be justified and derived explicitly, even if an existing package or function is applied. The quality of the presentation of results will be evaluated.
- (iv) You are free to choose the software for the assignment (Python, R, MATLAB, etc.). Excel is not suitable.
- (v) All codes used for the analysis must be **properly commented** and attached to the submission.
- (vi) The graded assignment will account for 50% of the final grade.

2 Dataset

Download historical prices of Apple Inc. from Yahoo Finance. Use daily historical returns from 2020 to 2026.

3 Description and Tasks

3.1 Description of the derivative

We consider a European floating-strike Asian call option whose payoff at maturity depends on the terminal stock price and on the average of the underlying asset prices observed over discrete times.

The payoff at maturity T is given by:

$$A_T = \max \left(S_n - \frac{1}{n+1} \sum_{t=0}^n S_t, 0 \right)$$

where:

- S_t is the asset price at time step $t = 0, \dots, n$, where n is the number of periods in the multi-period binomial model.
- $\frac{1}{n+1} \sum_{t=0}^n S_t$ is the arithmetic average of the stock prices over the life of the option.

3.2 Multi-period Binomial Tree Model Setup

Consider the multi-period Binomial Tree Model of Chapter 2, Slide 38, and set:

- (i) Maturity $T = 6$ months.
- (ii) Number of periods in the binomial tree is $n = 25$.
- (iii) Risk-free rate is 1% per annum.
- (iv) Initial price S_0 is the price of Apple Inc. on 28 April 2026.
- (v) Up and down factors:

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}},$$

where $\Delta t = \frac{T}{n}$ and σ is the **annualized volatility** of the stock return. We assume

$$\sigma = \sigma_{\text{daily}} \sqrt{250},$$

where σ_{daily} is the sample standard deviation of daily returns.

- (vi) Probability $p = \frac{1}{2}$.

Remark: We have

$$\log \left(\frac{S_{t+\Delta t}}{S_t} \right) = \begin{cases} \sigma\sqrt{\Delta t} & \text{with probability } p = \frac{1}{2} \\ -\sigma\sqrt{\Delta t} & \text{with probability } p = \frac{1}{2} \end{cases}$$

It follows that:

$$\text{Var} \left(\log \left(\frac{S_{t+\Delta t}}{S_t} \right) \right) = \frac{1}{2} \sigma^2 \Delta t + \frac{1}{2} \sigma^2 \Delta t = \sigma^2 \Delta t.$$

Under the assumption that returns are independent, we then have:

$$\text{Var} \left(\log \left(\frac{S_n}{S_0} \right) \right) = \sigma^2 T.$$

3.3 Tasks

- (i) Build the binomial tree for S_t , $t = 0, \dots, n$.
- (ii) For each terminal node of the tree, compute the arithmetic average stock price

$$\bar{S}_n = \frac{1}{n+1} \sum_{t=0}^n S_t.$$

- (iii) For each terminal node of the tree, compute the payoff of the floating-strike Asian call option:

$$A_T = \max(S_n - \bar{S}_n, 0).$$

- (iv) Compute the risk-neutral probabilities.
- (v) Apply backward induction to determine the arbitrage-free price of the option in each node of the tree. Report the price of the option at $t = 0$. Carefully explain why, for this derivative, the option value is not determined by the current stock price alone, but also by the path history summarized through the running average.
- (vi) Run a robustness check concerning the model assumptions relative to the interest rate and the volatility of stock returns.
- (vii) Use a normal approximation to approximate the distribution of

$$D_n = S_n - \bar{S}_n$$

and use this approximation to estimate the option price at time $t = 0$. Report and comment on the approximated price.

3.4 Additional guidance

In order to price the option by backward induction, you may find it useful to augment the state space and keep track, at each node, not only of the current stock price but also of the cumulative sum

$$C_t = \sum_{j=0}^t S_j.$$

At maturity, the payoff can then be written as

$$A_T = \max \left(S_n - \frac{C_n}{n+1}, 0 \right).$$

This makes clear that the derivative is **path dependent**. Your implementation should therefore explain how this additional state variable is handled.

For the normal approximation, derive and justify the mean and variance of the approximating random variable as carefully as possible. If you use simplifying assumptions, state them explicitly and comment on their implications.